Unit 3: Linear Functions

1. Write an equation of the line modeled in the table below.

x	1.5	4	6.5	7
y	10.25	13	15.75	16.3

$$\frac{13-10.25}{4-1.5} = \frac{2.75}{2.5} = 1.1$$

$$y - 10.25 = 1.1(x-1.5)$$

$$y - 10.25 = 1.1x - 1.65$$

$$+10.25 \qquad +10.25$$

$$y = 1.1x + 8.6$$

2. A line passes through the points $(6, 21)$ and $(13, -21)$. Find the equation of the line in slope intercept form.

$x_1 \; y_1 \qquad x_2 \; y_2$

$$\frac{-21-21}{13-6} = \frac{-42}{7} = -6$$

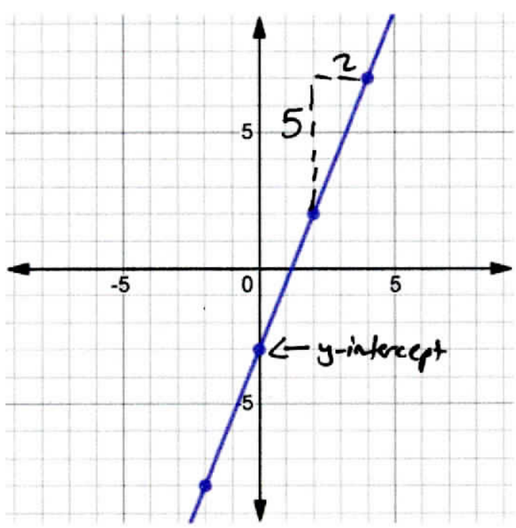
$$y - 21 = -6(x-6)$$

$$y - 21 = -6x + 36$$

$$+21 \qquad +21$$

$$y = -6x + 57$$

3. What is the equation of the line below?



$$y = \frac{5}{2}x - 3$$